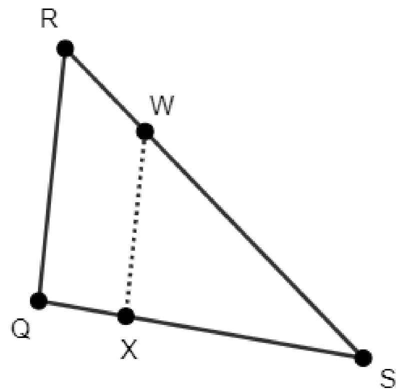
Geometry
Unit 6
Lesson 7 Practice

Name____________________
Date____________________
Period__________________

$\triangle QRS$ is shown, where $m\angle QRS = 50^\circ$, $m\angle RQS = 94^\circ$, and $m\angle RSQ = 36^\circ$. Point W is on $\overline{RS}$ and Point X is on $\overline{QS}$. $\overline{WX}$ is drawn as a dotted line.



1. Use a ruler and a protractor to complete the table.

Measurements of $\triangle QRS$		**Measurements of $\triangle XWS$**	
$QR =$	$m\angle QRS = 50^\circ$	$XW =$	$m\angle XWS =$
$RS =$	$m\angle RQS = 94^\circ$	$WS =$	$m\angle WXS = 94^\circ$
$QS =$	$m\angle RSQ = 36^\circ$	$XS =$	$m\angle WSX =$

2. The corresponding angles of $\triangle QRS$ and $\triangle XWS$ are _____________________.

3. Complete the table.

Sides	**Ratio**
Determine the ratio of $\overline{QR}$ to $\overline{XW}$.	
Determine the ratio of $\overline{RS}$ to $\overline{WS}$.	
Determine the ratio of $\overline{QS}$ to $\overline{XS}$.	

4. The corresponding sides of $\triangle QRS$ and $\triangle XWS$ are _____________________.

5. $\triangle QRS$ is similar to $\triangle$_________.

6. Describe a sequence of transformations that can be used to map $\triangle QRS$ onto $\triangle XWS$.

Complete each relationship for $\triangle QRS$ and $\triangle XWS$.

7. $\frac{RS}{} = \frac{}{XS}$

8. $\frac{WX}{} = \frac{}{QS}$

9. $\frac{RQ}{} = \frac{}{WS}$

10. Observing that $\frac{RQ}{WX} \neq \frac{RW}{WS}$, your classmate Charlee claims that $\triangle QRS$ and $\triangle XWS$ are not similar. If her logic is faulty, explain why.